

Code Sample

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This document contains excerpts from the replication code of a paper titled “Inequality in a Potential Equalizer of Opportunity: Variation in the Accessibility and Net Benefits of English-Medium Instruction in India.” Excerpts are followed by their corresponding outputs when appropriate. The project may have been updated since this file was generated; the most up-to-date versions of this document, the full paper, and its replication package are available online, and I would be happy to provide links for those interested. This PDF can be generated by running `make samples` or `scripts/run_full_build.sh` from the repo root.

Reconstructing District Boundary Transfers Between 1991 and 2001

A historical Census table reports how each 1991 district’s population was apportioned among 2001 districts after boundary changes. The file has no header, and long district names are split across continuation rows. The reader repairs only row patterns that match the table’s continuation structure, converts the transfer percentages to numbers, and requires the percentages assigned from every 1991 district to total 100% within 0.05 percentage points.

File: `R/io/read_inputs.R`

```
## Rejoin district names split across rows in the historical boundary table
##
## Rows with a blank 1991 population field are continuations. A trailing hyphen
## identifies a split 1991 district name; a 2001 name can continue on a row with
## no transfer percentages.
repair_district_carveout_wrapped_rows <- function(x) {
  out <- safe_df(x)
  if (!nrow(out)) return(out)
  blank <- function(value) is.na(value) || !nzchar(trimws(as.character(value)))
  drop <- rep(FALSE, nrow(out))
  for (i in seq_len(nrow(out))) {
```

```

if (i == 1L || !blank(out$pop_1991[[i]])) next
previous <- i - 1L
source_continuation <- !blank(out$district_1991[[i]])
target_continuation <- !blank(out$district_2001[[i]])
has_transfer <- !blank(out$pct_01in91[[i]]) || !blank(out$pct_91in01[[i]])

if (source_continuation) {
  previous_name <- plain_chr(out$district_1991[[previous]])
  if (!length(previous_name) || !grepl("-", trimws(previous_name[[1L]]))) {
    stop("Unexpected wrapped row in the district boundary table.", call. = FALSE)
  }
  continuation <- plain_chr(out$district_1991[[i]])[[1L]]
  combined <- join_district_carveout_wrapped_label(previous_name[[1L]], continuation)
  out$district_1991[[previous]] <- combined
  if (has_transfer) {
    out$district_1991[[i]] <- combined
    out$pop_1991[[i]] <- out$pop_1991[[previous]]
  }
}

if (!has_transfer && (source_continuation || target_continuation)) {
  if (target_continuation) {
    if (blank(out$district_2001[[previous]])) {
      stop("Unexpected continuation of a 2001 district name in the district boundary table.", call. =
        ↪ FALSE)
    }
    previous_target <- trimws(as.character(out$district_2001[[previous]]))
    continuation <- trimws(as.character(out$district_2001[[i]]))
    out$district_2001[[previous]] <- join_district_carveout_wrapped_label(
      previous_target, continuation
    )
  }
  drop[[i]] <- TRUE
}
out[!drop, , drop = FALSE]
}

district_carveout_rounding_tolerance_pp <- function() 0.05

#' Validate population shares in the 1991-to-2001 district transfer table

```

```

#'
validate_district_carveout_shares <- function(
  x,
  rounding_tolerance_pp = district_carveout_rounding_tolerance_pp()) {
  out <- safe_df(x)
  if (!nrow(out)) return(invisible(TRUE))
  share_cols <- c("pct_01in91", "pct_91in01")
  for (field in share_cols) {
    values <- num(out[[field]])
    bad <- is.finite(values) & (values < 0 | values > 100)
    if (any(bad)) {
      stop("District transfer shares must lie in [0, 100].", call. = FALSE)
    }
  }
  source_share <- num(out$pct_01in91)
  if (any(!is.finite(source_share))) {
    stop("Shares of each 1991 district assigned to 2001 districts must be complete.", call. = FALSE)
  }
  source_key <- paste(out$district_1991, out$pop_1991, sep = "__")
  source_sum <- vapply(split(source_share, source_key), sum, numeric(1))
  if (any(abs(source_sum - 100) > rounding_tolerance_pp)) {
    stop("Shares of each 1991 district assigned to 2001 districts must sum to 100 within rounding
    ↪ tolerance.", call. = FALSE)
  }
  invisible(TRUE)
}

read_district_carveouts <- function(path) {
  out <- utils::read.csv(
    path,
    header = FALSE,
    col.names = c("district_1991", "pop_1991", "district_2001", "pct_01in91", "pct_91in01"),
    stringsAsFactors = FALSE,
    na.strings = c("", "NA"),
    check.names = FALSE
  )
  out <- repair_district_carveout_wrapped_rows(out)
  out$district_1991 <- fill_down_missing(out$district_1991)
  out$pop_1991 <- num(gsub(",", "", fill_down_missing(out$pop_1991), fixed = TRUE))
  out$pct_01in91 <- num(out$pct_01in91)
  out$pct_91in01 <- num(out$pct_91in01)
}

```

```

    validate_district_carveout_shares(out)
    out
}

```

Reconstructing District Data from the 1991 Language Atlas

The 1991 Census Language Atlas repeats each district across eight language pages. Districts are identified by a Census state code plus a within-state district number, printed in the Atlas as a serial; some repeated rows omit it or contain errors from optical character recognition (OCR). The code fills a missing serial only when matching row labels agree, then validates the resulting district code against the official 1991 Primary Census Abstract population.

File: `scripts/build_language_atlas_1991.py`

```

def cross_page_serial_evidence(cells: list[dict[str, object]]) -> dict[tuple[int, str], tuple[int, ...]]:
    """Collect district serials printed with repeated Language Atlas row labels.

    The Atlas prints each district on eight language pages. The district serial is
    the within-state number paired with the Census state code. If one copy omits
    that number, repeated copies can supply it only when their printed serials agree.
    """
    observed: dict[tuple[int, str], set[int]] = {}
    unique_rows = {
        (int(cell["block"]), int(cell["page"]), int(cell["row_sequence"]),
         str(cell["row_label_raw"]).strip(), str(cell["serial_candidate"]).strip())
        for cell in cells
    }
    for block, _page, _sequence, label, serial_raw in unique_rows:
        if not label or label == "INDIA":
            continue
        serial = int(serial_raw) if serial_raw else parse_leading_serial(label)
        if serial is not None:
            observed.setdefault((block, label), set()).add(serial)
    return {key: tuple(sorted(values)) for key, values in observed.items()}

def reconcile_district_serial(
    cell: dict[str, object],

```

```

    evidence: dict[tuple[int, str], tuple[int, ...]],
) -> tuple[int | None, str, str]:
    """Choose a district serial from the row itself or agreeing repeated copies."""
    label = str(cell["row_label_raw"]).strip()
    serial_column = int(cell["serial_candidate"]) if str(cell["serial_candidate"]).strip() else None
    serial_label = parse_leading_serial(label)
    repeated = evidence.get((int(cell["block"]), label), ())
    repeated_text = ";".join(map(str, repeated))

    if serial_column is not None and serial_label is not None and serial_column != serial_label:
        return None, "direct_conflict", repeated_text
    direct = serial_column if serial_column is not None else serial_label
    direct_source = "serial_column" if serial_column is not None else "label_prefix"

    if direct is not None:
        # Prefer a serial printed on the current row. Use repeated copies of the
        # same label only when the current row has no serial because continuation
        # pages contain occasional OCR errors.
        return direct, direct_source, repeated_text
    if len(repeated) == 1:
        return repeated[0], "cross_page_exact_label", repeated_text
    if len(repeated) > 1:
        return None, "cross_page_conflict", repeated_text
    return None, "missing", repeated_text

def build_district_population_validation(
    cells: list[dict[str, object]],
    state_crosswalk: dict[str, dict[str, str]],
    pca91: dict[tuple[str, str], int],
) -> list[dict[str, object]]:
    rows = merge_population_continuations(cells, state_crosswalk)
    serial_evidence = cross_page_serial_evidence(cells)
    seen_state_codes: set[str] = set()
    current_state: dict[str, str] | None = None
    candidates: list[dict[str, object]] = []
    for cell in rows:
        label = str(cell["row_label_raw"]).strip()
        if label == "INDIA":
            continue
        mapped_state = state_crosswalk.get(label)

```

```

if mapped_state is not None and mapped_state["state_code_1991"] not in seen_state_codes:
    current_state = mapped_state
    seen_state_codes.add(mapped_state["state_code_1991"])
    continue

serial, serial_source, cross_page_serials = reconcile_district_serial(cell, serial_evidence)

state_code = current_state["state_code_1991"] if current_state else ""
state_name = current_state["state_name_1991"] if current_state else ""
district_code = f"{serial:02d}" if serial is not None else ""
pca_population = pca91.get((state_code, district_code)) if state_code and district_code else None
atlas_population = (
    int(cell["speaker_count_candidate"])
    if str(cell["speaker_count_candidate"]).strip()
    else None
)
if atlas_population is not None and pca_population is not None and pca_population > 0:
    abs_diff = abs(atlas_population - pca_population)
    rel_diff = abs_diff / pca_population
else:
    abs_diff = None
    rel_diff = None
candidates.append({
    "block": cell["block"],
    "page": cell["page"],
    "row_sequence": cell["row_sequence"],
    "row_sequence_end": cell["row_sequence_end"],
    "row_merge_status": cell["row_merge_status"],
    "row_label_raw": label,
    "state_code_1991": state_code,
    "state_name_1991": state_name,
    "district_code_1991": district_code,
    "serial_source": serial_source,
    "cross_page_serials": cross_page_serials,
    "atlas_population_raw": cell["raw_value"],
    "atlas_population_candidate": "" if atlas_population is None else atlas_population,
    "pca91_population": "" if pca_population is None else pca_population,
    "population_abs_diff": "" if abs_diff is None else abs_diff,
    "population_relative_diff": "" if rel_diff is None else round(rel_diff, 8),
    "identity_status": "",
    "population_status": "",

```

```

        "promotion_status": "review_required",
    })

expected_state_codes = {row["state_code_1991"] for row in state_crosswalk.values()}
if seen_state_codes != expected_state_codes:
    missing = sorted(expected_state_codes - seen_state_codes)
    extra = sorted(seen_state_codes - expected_state_codes)
    raise ValueError(
        "Language Atlas state headings do not match the expected 1991 states; "
        f"missing={missing}; extra={extra}"
    )
pca_state_codes = {state_code for state_code, _ in pca91}
if pca_state_codes != expected_state_codes:
    raise ValueError("Language Atlas state names and 1991 Primary Census Abstract state codes disagree")

key_counts: dict[tuple[str, str], int] = {}
for row in candidates:
    key = (str(row["state_code_1991"]), str(row["district_code_1991"]))
    if all(key):
        key_counts[key] = key_counts.get(key, 0) + 1

for row in candidates:
    key = (str(row["state_code_1991"]), str(row["district_code_1991"]))
    if not row["state_code_1991"]:
        identity_status = "missing_state_context"
    elif row["serial_source"] in {"direct_conflict", "cross_page_conflict"}:
        identity_status = "serial_conflict"
    elif not row["district_code_1991"]:
        identity_status = "missing_district_serial"
    elif key_counts.get(key, 0) > 1:
        identity_status = "duplicate_district_code"
    elif key not in pca91:
        identity_status = "missing_pca_district"
    else:
        identity_status = "candidate_official_code"
    row["identity_status"] = identity_status

    if row["atlas_population_candidate"] == "":
        population_status = "atlas_population_unparsed"
    elif row["pca91_population"] == "":
        population_status = "pca_population_missing"

```

```

elif float(row["population_relative_diff"]) <= POPULATION_RELATIVE_TOLERANCE:
    population_status = "within_1pct"
else:
    population_status = "over_1pct"
row["population_status"] = population_status
if identity_status == "candidate_official_code" and population_status == "within_1pct":
    row["promotion_status"] = "population_validated_candidate"
return candidates

```

Output: outputs/figures/appendix/appendix_historical_language_persistence.pdf

Rendering code: R/output/appendix_data_construction_exhibits.R and R/output/save_figures.R

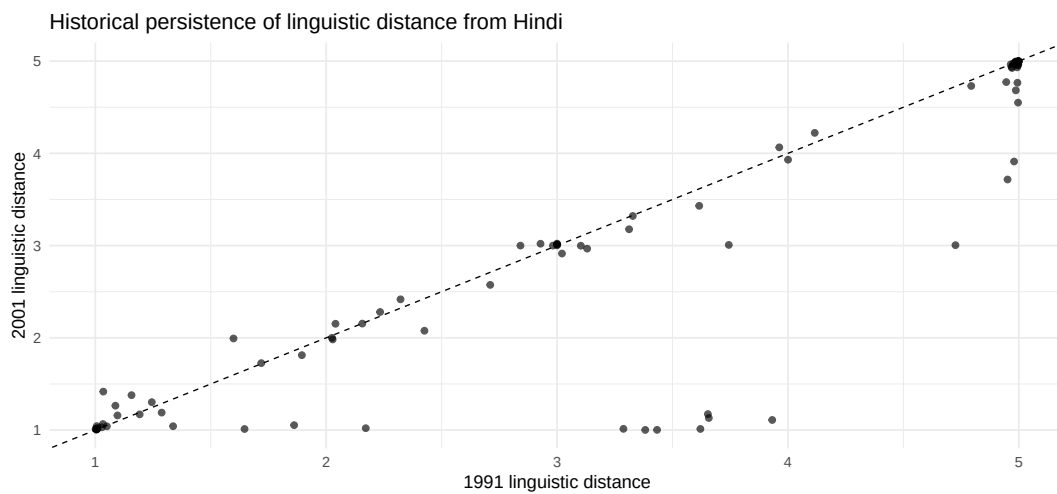


Figure 2: Persistence of speaker-weighted linguistic distance from Hindi from 1991 to 2001. Each point is a 2001 Census district with a reconstructed measure in both years; the dashed line is the 45-degree line.

Sources: Census of India 1991 Language Atlas (Office of the Registrar General and Census Commissioner, India 2004) and Census of India 2001 language tables (Office of the Registrar General and Census Commissioner of India 2001).

Constructing the NSS Survey Design

The 2007–08 National Sample Survey (NSS) education survey samples households within first-stage units such as villages or urban blocks. This function combines the state, rural/urban sector, stratum, and substratum fields into NSS survey strata, attaches household sampling weights, and constructs the `survey::svydesign()` specification used for the enrollment probit. Final estimation stops if a required design field is missing.

File: R/selection/estimate_selection_probit.R

```

# List NSS fields needed to reproduce the survey sample design
#
selection_survey_design_variables <- function() {
  c("FSU_SL_NO", "weight", "STATE", "SECTOR", "STRATUM", "SUB_STRATUM_NO")
}

# Construct the NSS design used for weighted enrollment estimation
#
build_survey_design_selection <- function(selection_df, require_all = FALSE) {
  required <- selection_survey_design_variables()
  missing <- setdiff(required, names(selection_df))
  if (isTRUE(require_all) && length(missing)) {
    stop(
      "NSS survey design is missing required fields: ",
      paste(missing, collapse = ", "),
      call. = FALSE
    )
  }

  psu <- first_col(selection_df, c("FSU_SL_NO", "fsu", "PSU", "psu"))
  weight <- first_col(selection_df, c("weight", "WEIGHT", "Multiplier", "multiplier"))
  strata_cols <- intersect(c("STATE", "SECTOR", "STRATUM", "SUB_STRATUM_NO"), names(selection_df))
  if (is.null(psu) || is.null(weight) || !length(strata_cols)) {
    if (isTRUE(require_all)) {
      stop("The required NSS survey design could not be constructed.", call. = FALSE)
    }
    return(NULL)
  }

  # NSS strata are defined by the joint state, rural/urban sector, stratum, and
  # substratum identifiers used in the survey files.
  selection_df$.survey_strata <- interaction(selection_df[strata_cols], drop = TRUE)
  survey::svydesign(
    ids = stats::as.formula(paste0("~", psu)),
    strata = ~.survey_strata,
    weights = stats::as.formula(paste0("~", weight)),
    data = selection_df,
    nest = TRUE
  )
}

```

Average Marginal Effects on the Estimation Sample

The code extracts the exact observations and NSS sampling weights used to fit the enrollment probit, checks that the weights align with those observations, and passes both to `marginaleffects::avg_slopes()`. The result is an average change in predicted enrollment on the probability scale over the survey-weighted estimation sample.

File: R/selection/compute_average_marginal_effects.R

```
## Recover the observations and sampling weights used to fit the enrollment model
##
ame_model_data_and_weights <- function(model) {
  model_data <- as.data.frame(stats::model.frame(model))

  model_weights <- if (inherits(model, "svyglm") && !is.null(model$survey.design)) {
    stats::weights(model$survey.design, type = "sampling")
  } else {
    tryCatch(stats::weights(model, type = "prior"), error = function(e) NULL)
  }

  if (is.null(model_weights)) model_weights <- rep(1, nrow(model_data))
  model_weights <- as.numeric(model_weights)
  if (length(model_weights) != nrow(model_data)) {
    stop("AME averaging weights do not align with the estimation sample.", call. = FALSE)
  }
  if (any(!is.finite(model_weights)) || any(model_weights < 0) || sum(model_weights) <= 0) {
    stop("AME averaging weights must be finite, nonnegative, and have positive total weight.", call. = FALSE)
  }

  list(data = model_data, wts = model_weights)
}

## Compute average marginal effects over the fitted NSS sample
##
compute_ames_marginaleffects <- function(model) {
  # Pass the fitted rows and sampling weights explicitly so avg_slopes() averages
# predicted changes over the same weighted observations used for estimation after
# the fitted model has been saved and reloaded. Uncertainty comes from the
# survey-weighted model's covariance matrix.
  amed <- ame_model_data_and_weights(model)
  marginaleffects::avg_slopes(
    model,
```

```

newdata = amed$data,
wts = amed$wts,
vcov = TRUE,
type = "response"
)
}

```

Output: outputs/tables/appendix/appendix_selection_ame.tex

Rendering code: R/output/appendix_selection_exhibits.R and R/output/save_tables.R

Table 7: Average Marginal Effects for School Enrollment

Term	Enrolled (1 = yes)
Age (years)	-0.028*** (0.000)
Female (ref: Male)	0.044*** (0.003)
Household size	-0.004*** (0.001)
Religion: Muslim (ref: Hindu)	-0.126*** (0.007)
Religion: Christian	-0.043*** (0.013)
Religion: Sikh	0.040 (0.026)
Religion: Jain	-0.001 (0.012)
Religion: Buddhist	0.047* (0.024)
Religion: Zoroastrian	0.101*** (0.034)
Religion: Other	0.252*** (0.003)
Social group: Scheduled Tribe (ref: Other)	-0.102*** (0.008)
Social group: Scheduled Caste	-0.075*** (0.006)
Social group: Other Backward Class	-0.029*** (0.005)
Urban (ref: Rural)	-0.003 (0.005)
Distance 1–2km (ref: <1km)	-0.077*** (0.026)
Distance 2–3km	-0.001 (0.010)
Distance 3–5km	0.004 (0.066)
Distance > 5km	0.036 (0.035)
Male household education: Literate, no school (ref: Illiterate)	0.097*** (0.017)
Male household education: Literate, school < primary	0.127*** (0.006)

Table 7: Average Marginal Effects for School Enrollment (*continued*)

Term	Enrolled (1 = yes)
Male household education: Primary	0.142*** (0.005)
Male household education: Upper primary	0.195*** (0.006)
Male household education: Secondary	0.212*** (0.007)
Male household education: Higher secondary	0.247*** (0.008)
Male household education: Postsecondary+	0.245*** (0.009)

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Entries are average changes in predicted enrollment probability. Continuous covariates are average slopes; categorical covariates are average discrete comparisons against the stated reference category, evaluated over the observed covariate distribution. Standard errors use the NSS sampling design, including household weights, primary sampling units, and strata formed from state, stratum, and substratum identifiers. The model also controls for district averages of schooling costs and benefits among enrolled children; the table displays marginal effects for child and household covariates. Source: NSS 64th Round, 2007–08, Participation and Expenditure in Education.

Assessing Weak-Instrument Strength with the Montiel Olea–Pflueger Effective F

The paper uses linguistic distance from Hindi as an instrument for district exposure to English-medium schooling. This function evaluates that first stage with the Montiel Olea–Pflueger effective F. It recovers the regressors and instruments from the fitted `ivreg` model, uses the same estimation rows, and calls `momentfit::MOPtest()` with state-clustered covariance when state identifiers are available. The table pairs this strength measure with two-stage least squares (2SLS) and Anderson–Rubin results.

File: R/iv/weak_identification.R

```
mop_effective_f <- function(
  model, analysis_data, tau = 0.10, size = 0.05) {
  unavailable <- function(reason) {
    list(
      statistic = NA_real_, critical_value = NA_real_, p.value = NA_real_,
      effective_df = NA_real_, tau = tau, size = size,
      status = "not_estimated", reason = reason
    )
  }

  if (!inherits(model, "ivreg")) {
    return(unavailable("Montiel Olea-Pflueger effective F requires an ivreg model."))
  }
}
```

```

if (!requireNamespace("momentfit", quietly = TRUE)) {
  return(unavailable("Package 'momentfit' is not installed."))
}

if (!is.finite(tau) || tau <= 0 || !is.finite(size) || size <= 0 || size >= 1) {
  return(unavailable("Montiel Olea-Pflueger tau and size must be valid probabilities."))
}

endogenous <- suppressWarnings(as.integer(model$endogenous))
if (length(endogenous) != 1L || !is.finite(endogenous)) {
  return(unavailable("Montiel Olea-Pflueger effective F requires exactly one endogenous regressor."))
}

regression_formula <- tryCatch(
  stats::formula(model, component = "regressors"),
  error = function(e) NULL
)
instrument_formula <- tryCatch(
  stats::formula(stats::delete.response(stats::terms(model, component = "instruments")),
  error = function(e) NULL
)
if (is.null(regression_formula) || is.null(instrument_formula)) {
  return(unavailable("Could not recover ivreg component formulas for Montiel Olea-Pflueger effective F."))
}

needed <- unique(c(all.vars(regression_formula), all.vars(instrument_formula)))
data <- iv_analysis_frame(analysis_data, needed)
missing <- setdiff(needed, names(data))
if (length(missing)) {
  return(unavailable(paste(
    "Missing variables for Montiel Olea-Pflueger effective F:",
    paste(missing, collapse = ", ")
  )))
}

moment_data <- data[needed]
if (any(!stats::complete.cases(moment_data))) {
  return(unavailable("Regressors or instruments used by the fitted IV model contain missing values."))
}

stored_cluster <- attr(model, "cluster_state", exact = TRUE)
cluster <- if (
  length(stored_cluster) == stats::nobs(model) &&
  !anyNA(stored_cluster) && length(unique(stored_cluster)) >= 2L

```

```

) {
  as.vector(stored_cluster)
} else {
  iv_model_cluster(model, data)
}

covariance <- if (is.null(cluster)) "MDS" else "CL"
covariance_options <- if (is.null(cluster)) {
  list(type = "HCO")
} else {
  # MOPtest() uses clustered HCO moment covariance with its finite-cluster
  # adjustment here. The separately reported first-stage Wald F uses HC1, so the
  # two statistics follow different covariance conventions.
  list(
    cluster = data.frame(cluster = cluster),
    type = "HCO", cadjust = TRUE, multi0 = FALSE
  )
}

result <- tryCatch({
  moment_model <- momentfit::momentModel(
    regression_formula,
    instrument_formula,
    data = moment_data,
    vcov = covariance,
    vcovOptions = covariance_options
  )
  momentfit::MOPtest(
    moment_model,
    tau = tau,
    size = size,
    estMethod = "TSLS",
    simplified = TRUE,
    print = FALSE
  )
}, error = function(e) e)
if (inherits(result, "error")) return(unavailable(conditionMessage(result)))

values <- suppressWarnings(as.numeric(result[c("Feff", "critValue", "pvalue", "Keff"])))
names(values) <- c("statistic", "critical_value", "p.value", "effective_df")
if (length(values) != 4L || any(!is.finite(values))) {
  return(unavailable("momentfit::MOPtest() did not return finite effective-F statistics."))
}

```

```
}  
  
c(  
  as.list(values),  
  list(tau = tau, size = size, status = "estimated", reason = NA_character_)  
)  
}
```

Output: outputs/tables/appendix/appendix_iv_weak_inference.tex

Rendering code: R/output/appendix_identification_exhibits.R and R/output/save_tables.R

Table 10: Inference with Weak Instruments for Long-Run Consumption Change

	Real log consumption change, 2004-05 to 2022-23	Real log consumption change, 2004-05 to 2023-24
EMI exposure	0.117 (0.182)	0.137 (0.281)
MOP effective F	0.69	0.41
AR p -value at zero	0.029	0.081
AR 95% accepted set	$[-0.193, -0.030]^{\dagger} \cup [0.009, 0.193]^{\dagger}$	$[-0.186, 0.186]^{\dagger}$
Direct effect threshold for $\beta = 0$ (share of reduced form)	16.6%	0.0%
Observations	524	522

The table reports conventional 2SLS coefficients with state-clustered standard errors. Instrument strength is summarized by the Montiel Olea–Pflueger effective F statistic, and Anderson–Rubin tests provide inference robust to weak instruments. The 95% AR accepted sets are displayed as unions of closed intervals. A dagger marks a component that reaches an evaluated search boundary. Its outer endpoint is the numerical search limit and should not be interpreted as a finite confidence bound. The direct effect threshold is the smallest permitted direct effect, expressed as a share of the absolute reduced form, for which the AR set includes $\beta = 0$. Sources: Census of India 2001; NSS 61st Round, 2004–05; NSS 64th Round, 2007–08; HCES 2022–23; HCES 2023–24.

Mapping District Estimates on 2001 Census Boundaries

To display estimates on a common geography, the code joins analysis values to every 2001 Census district polygon before plotting. Districts outside an estimation sample remain visible with a no-data fill, disputed territories receive a separate no-data treatment, and state borders are drawn from the full district geometry. The plotting function handles both categorical and continuous measures; continuous measures can use diverging color scales.

File: R/output/save_figures.R

```
build_public_ggplot_map <- function(plot_data, spec, boundary_reference = NULL) {
  need_pkg("ggplot2", "classified choropleth maps")
  style <- public_map_style(spec$variable)
  fill <- public_map_fill(plot_data, spec$variable, style)
  plot_data <- fill$data
  overlay <- plot_data[map_overlay_rows(plot_data, fill$fill), , drop = FALSE]
  if (!nrow(overlay)) {
    stop("Map figure '", spec$name, "' has no district values to plot for variable '", spec$variable, "'",
      ↪ call. = FALSE)
  }

  disputed_display <- public_map_disputed_display(plot_data, boundary_reference)
  display_data <- mask_public_map_disputed_areas(plot_data, disputed_display)
```

```

state_outlines <- public_map_state_outlines(display_data)
base <- ggplot2::ggplot() +
  ggplot2::geom_sf(
    data = display_data, ggplot2::aes(fill = .data[[fill$fill]]),
    color = "grey55", linewidth = map_district_boundary_linewidth(),
    linetype = "solid"
  )
if (!is.null(disputed_display) && nrow(disputed_display)) {
  base <- base + ggplot2::geom_sf(
    data = disputed_display, fill = map_disputed_no_data_colour(),
    color = "grey45", linewidth = map_disputed_boundary_linewidth(),
    linetype = "solid"
  )
}
if (!is.null(state_outlines) && nrow(state_outlines)) {
  base <- base + ggplot2::geom_sf(
    data = state_outlines, color = "grey15",
    linewidth = map_major_boundary_linewidth(), linetype = "solid"
  )
}
base <- base +
  ggplot2::coord_sf(datum = NA) +
  ggplot2::labs(fill = fill$title) +
  ggplot2::theme_void(base_size = 10) +
  ggplot2::theme(
    legend.position = "right",
    legend.title = ggplot2::element_text(size = 12, face = "bold"),
    legend.text = ggplot2::element_text(size = 10),
    legend.key.height = grid::unit(16, "pt"),
    legend.spacing.y = grid::unit(4, "pt"),
    plot.margin = grid::unit(c(2, 2, 2, 2), "pt")
  )

if (isTRUE(fill$continuous)) {
  return(base + ggplot2::scale_fill_gradientn(
    colours = fill$colors,
    limits = fill$limits,
    na.value = map_no_data_colour(),
    guide = public_map_colorbar_guide()
  ))
}

```

```

base +
  ggplot2::scale_fill_manual(
    values = fill$colors,
    breaks = names(fill$colors),
    limits = names(fill$colors),
    drop = FALSE,
    na.translate = TRUE,
    na.value = map_no_data_colour()
  ) +
  ggplot2::guides(
    fill = ggplot2::guide_legend(
      override.aes = map_legend_override(fill$colors)
    )
  )
}

complete_public_map_geometry <- function(district_panel, map_geometry) {
  if (!has_sf_geometry(map_geometry)) {
    stop("District maps require the complete 2001 Census district geometry.", call. = FALSE)
  }
  key <- "target_unit_2001"
  if (!key %in% names(map_geometry) && "unit_id" %in% names(map_geometry)) {
    map_geometry[[key]] <- map_geometry$unit_id
  }
  if (!key %in% names(map_geometry) || !key %in% names(district_panel)) {
    stop("Map geometry and analysis data must contain a 2001 Census district key.", call. = FALSE)
  }
  if (anyDuplicated(map_geometry[[key]])) {
    stop("Complete 2001 Census map geometry must contain one row per district.", call. = FALSE)
  }
  attributes <- if (inherits(district_panel, "sf")) sf::st_drop_geometry(district_panel) else
  ↪ safe_df(district_panel)
  attributes <- attributes[!duplicated(attributes[[key]]), , drop = FALSE]
  out <- merge(map_geometry, attributes, by = key, all.x = TRUE, sort = FALSE)

  # Read the state code from each 2001 Census district identifier. This preserves
  # complete state and national borders when a district has no value to plot.
  boundary_state <- public_map_state_code_2001(out[[key]])
  if ("state_code_2001" %in% names(out)) {
    panel_state <- plain_chr(out$state_code_2001)
  }
}

```

```

mismatch <- !is.na(panel_state) & nzchar(panel_state) & panel_state != boundary_state
if (any(mismatch)) {
  stop("Analysis state codes disagree with the state codes in the 2001 Census district identifiers.",
    ↪ call. = FALSE)
}
}
out$state_code_2001 <- boundary_state
out
}

save_map_plot_formats <- function(map_plot, path_base, formats, width = 8, height = 6, dpi = 300) {
  save_plot_formats(map_plot, path_base, formats, width = width, height = height, dpi = dpi)
}

save_map_figure <- function(spec, path_base, district_panel, map_geometry, boundary_reference, formats) {
  if (!has_sf_geometry(district_panel)) {
    stop("Map figure '", spec$name, "' requires an sf district_panel with validated geometry.", call. =
      ↪ FALSE)
  }
  if (is.null(spec$variable) || !spec$variable %in% names(district_panel)) {
    stop("Map figure '", spec$name, "' is missing variable '", spec$variable, "'.", call. = FALSE)
  }

  plot_data <- complete_public_map_geometry(district_panel, map_geometry)
  plot_data <- prepare_public_map_data(plot_data, spec$variable)
  p <- build_public_ggplot_map(plot_data, spec, boundary_reference = boundary_reference)
  save_map_plot_formats(p, path_base, formats, width = 7.2, height = 5.2, dpi = 300)
}

```

Outputs: outputs/figures/main/map_emi_share_enrolled.pdf, outputs/figures/main/map_emi_exposure.pdf, outputs/figures/main/map_public_emi_exposure.pdf, and outputs/figures/main/map_private_emi_exposure.pdf

Rendering code: R/output/make_figures.R

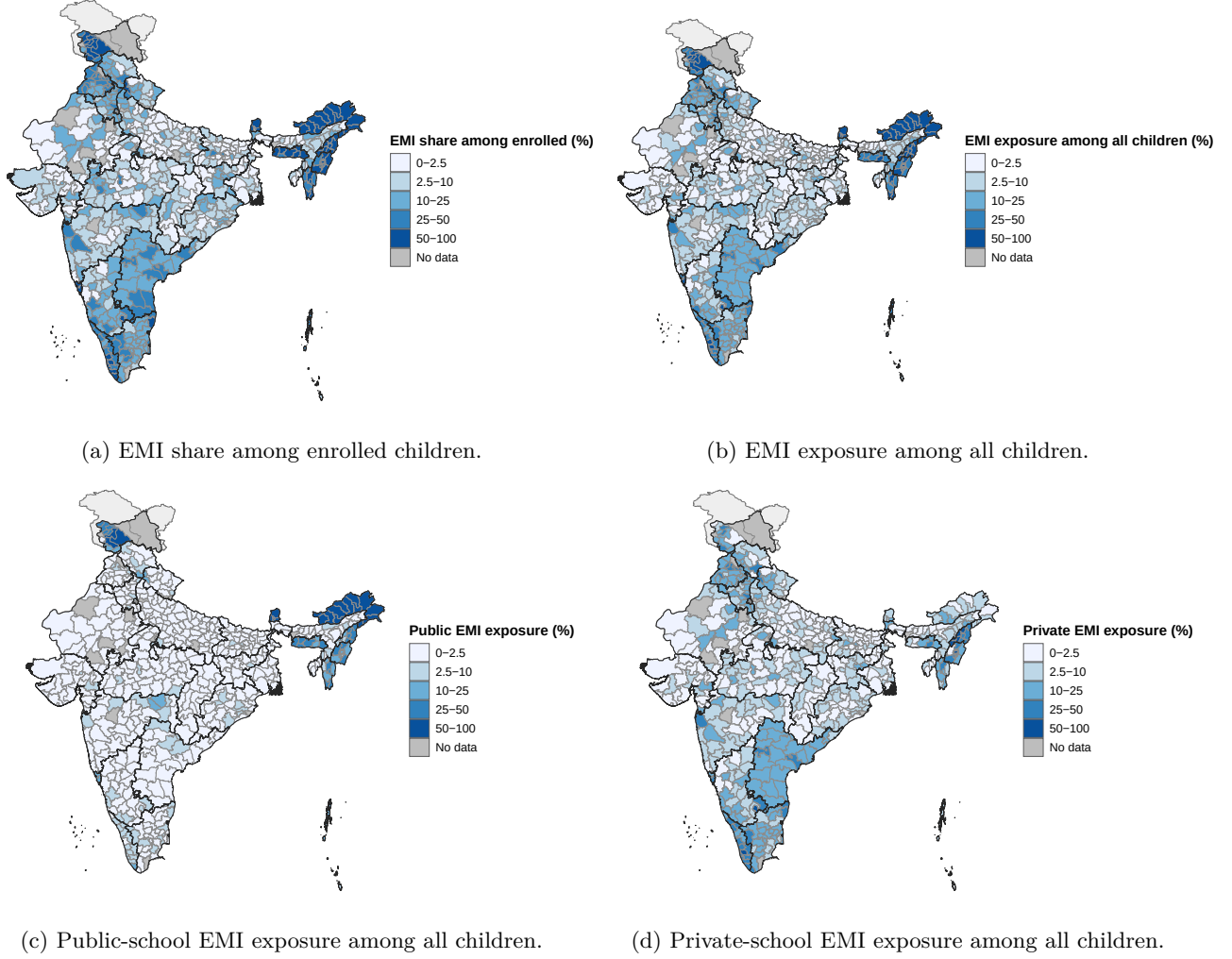


Figure 3: District distribution of English-medium schooling measures on 2001 Census boundaries. All panels use a common percentage scale. Source: NSS 64th Round, 2007–08 (National Sample Survey Office 2011). TENGshe (2014) and Natural Earth (2022) provide geospatial data. District boundaries reflect 2001 geography. Politically disputed regions are identified using Natural Earth, and those without data—Aksai Chin, Azad Kashmir, Gilgit-Baltistan, the Trans-Karakoram Tract, and Siachen Glacier—are colored light gray. Their inclusion does not imply a position on territorial disputes.

Linking Rural and Urban State CPI Series Across Index Bases

The historical Consumer Price Index for Rural Labourers (CPI-RL) and Consumer Price Index for Industrial Workers (CPI-IW) use different index bases from the later state rural/urban CPI (CPI-R/U). For each state and rural/urban sector, the code rescales the pre-2013 index by the median monthly ratio of CPI-R/U to the historical index over January 2013–December 2014. It requires at least six overlap months and records both the overlap count and the dispersion of the monthly ratios.

```

build_temporal_price_series <- function(
  price_sources,
  switch_date = as.Date("2013-01-01"),
  overlap_start = as.Date("2013-01-01"),
  overlap_end = as.Date("2014-12-01"),
  minimum_link_months = 6L,
  pre_switch_start = NULL,
  pre_switch_end = NULL) {
  switch_date <- price_boundary(switch_date)
  old <- select_pre_2013_price_series(price_sources)
  new <- select_post_2013_price_series(price_sources)
  links <- summarise_price_links(old, new, overlap_start, overlap_end)

  minimum_link_months <- as.integer(minimum_link_months)
  if (length(minimum_link_months) != 1L || is.na(minimum_link_months) || minimum_link_months < 1L) {
    stop("minimum_link_months must be one positive integer.", call. = FALSE)
  }

  old_groups <- unique(old[old$period < switch_date, c("state_code", "sector"), drop = FALSE])
  linked_groups <- links[links$link_months >= minimum_link_months, c("state_code", "sector"), drop = FALSE]
  linked_key <- paste(linked_groups$state_code, linked_groups$sector, sep = "\r")
  missing_key <- !paste(old_groups$state_code, old_groups$sector, sep = "\r") %in% linked_key
  if (any(missing_key)) {
    bad <- old_groups[missing_key, , drop = FALSE]
    stop(
      "Pre-2013 price series have fewer than the required overlap months with CPI-R/U: ",
      paste(paste(bad$state_code, bad$sector, sep = "/"), collapse = ", "),
      call. = FALSE
    )
  }

  old <- merge(old, links, by = c("state_code", "sector"), all.x = TRUE, sort = FALSE)
  old <- old[old$period < switch_date, , drop = FALSE]
  if (!is.null(pre_switch_start) || !is.null(pre_switch_end)) {
    if (is.null(pre_switch_start) || is.null(pre_switch_end)) {
      stop("Both pre_switch_start and pre_switch_end are required when trimming the old price series.",
        call. = FALSE)
    }
    pre_switch_start <- price_boundary(pre_switch_start)
  }

```

```

pre_switch_end <- price_boundary(pre_switch_end)
if (pre_switch_end < pre_switch_start || pre_switch_end >= switch_date) {
  stop("The requested pre-2013 period must end before the CPI switch date.", call. = FALSE)
}
old <- old[old$period >= pre_switch_start & old$period <= pre_switch_end, , drop = FALSE]
}
old$index_unlinked <- old$index
# Put each historical CPI series on the CPI-R/U scale before appending the
# post-2013 observations.
old$index <- num(old$index) * num(old$link_factor)

new <- new[new$period >= switch_date, , drop = FALSE]
new$link_factor <- 1
new$link_months <- NA_integer_
new$link_ratio_mad <- NA_real_
new$link_ratio_min <- NA_real_
new$link_ratio_max <- NA_real_
new$index_unlinked <- new$index

columns <- c(
  "state_code", "sector", "period", "index", "index_unlinked", "price_source",
  "link_factor", "link_months", "link_ratio_mad", "link_ratio_min", "link_ratio_max"
)
index <- rbind(old[columns], new[columns])
index$year <- as.integer(format(index$period, "%Y"))
index$month <- as.integer(format(index$period, "%m"))
index <- index[order(index$state_code, index$sector, index$period), , drop = FALSE]
rownames(index) <- NULL
validate_temporal_price_chain(index, switch_date)

structure(
  list(index = index, links = links, switch_date = switch_date),
  class = "emi_temporal_price_series"
)
}

```

Rural and Urban State Price Deflators for Comparable Real Consumption

Linking the CPI series accounts for price change over time. Comparing consumption across states also requires a rural/urban state price adjustment. The Tendulkar poverty lines provide separate rupee thresholds for each state and rural/urban sector; dividing them by the all-India rural reference yields the cross-state price relatives. The

code combines those relatives with the linked CPI and expresses household consumption in comparable 2011–12 rupees. The figure below compares district consumption estimates from the 2022–23 and 2023–24 Household Consumption Expenditure Survey (HCES) rounds after this adjustment.

File: R/prices/price_deflators.R

```
build_state_sector_deflator <- function(
  temporal_index, spatial_relatives, reference_period = NULL, reference_index = NULL) {
  idx <- safe_df(temporal_index)
  spatial <- safe_df(spatial_relatives)
  validate_price_index(idx, keys = c("state_code", "sector", "period"))
  required_spatial <- c("state_code", "sector", "spatial_price_relative")
  missing <- setdiff(required_spatial, names(spatial))
  if (length(missing)) {
    stop("Spatial price relatives are missing columns: ", paste(missing, collapse = ", "), call. = FALSE)
  }
  if (anyDuplicated(spatial[c("state_code", "sector")])) {
    stop("Spatial price relatives must have one row per state and sector.", call. = FALSE)
  }
  if (any(!positive_finite(num(spatial$spatial_price_relative)))) {
    stop("Spatial price relatives must be positive and finite.", call. = FALSE)
  }

  if (is.null(reference_index)) {
    if (is.null(reference_period)) stop("A price reference period or reference-index table is required.",
    ↪ call. = FALSE)
    reference_period <- as.Date(reference_period)
    ref <- idx[idx$period %in% reference_period, , drop = FALSE]
    if (!nrow(ref)) stop("No observations fall in the requested price reference period.", call. = FALSE)
    ref_mean <- stats::aggregate(index ~ state_code + sector, ref, function(x) mean(num(x), na.rm = TRUE))
    names(ref_mean)[names(ref_mean) == "index"] <- "reference_index"
  } else {
    ref_mean <- safe_df(reference_index)
    required_reference <- c("state_code", "sector", "reference_index")
    missing_reference <- setdiff(required_reference, names(ref_mean))
    if (length(missing_reference)) {
      stop("Reference-price table is missing columns: ", paste(missing_reference, collapse = ", "), call. =
      ↪ FALSE)
    }
    if (anyDuplicated(ref_mean[c("state_code", "sector")])) ||
    ↪ any(!positive_finite(ref_mean$reference_index))) {
```

```

    stop("Reference-price table must have one positive row per state and rural/urban sector.", call. =
      ↪ FALSE)
  }

  ref_mean <- ref_mean[required_reference]
}

out <- merge(
  idx,
  spatial,
  by = c("state_code", "sector"), all.x = TRUE, sort = FALSE
)

out <- merge(out, ref_mean, by = c("state_code", "sector"), all.x = TRUE, sort = FALSE)
if ("source_state_code" %in% names(out) && any(!positive_finite(out$reference_index))) {
  donor_reference <- ref_mean
  names(donor_reference)[names(donor_reference) == "state_code"] <- "source_state_code"
  names(donor_reference)[names(donor_reference) == "reference_index"] <- "donor_reference_index"
  out <- merge(out, donor_reference, by = c("source_state_code", "sector"), all.x = TRUE, sort = FALSE)
  missing_reference <- !positive_finite(out$reference_index)
  out$reference_index[missing_reference] <- out$donor_reference_index[missing_reference]
  out$donor_reference_index <- NULL
}

# The product adjusts for the rural/urban price level in each state and for
# price change from the reference period to the survey month.
out$temporal_price_relative <- num(out$index) / num(out$reference_index)
out$price_deflator <- num(out$spatial_price_relative) * num(out$temporal_price_relative)
if (any(!positive_finite(out$price_deflator))) {
  stop("Price deflators must be positive and finite for every retained state, sector, and month.", call. =
    ↪ FALSE)
}

out <- out[order(out$state_code, out$sector, out$period), , drop = FALSE]
rownames(out) <- NULL
out
}

build_state_sector_price_deflators <- function(
  temporal_series,
  state_rules = read_price_state_crosswalk(),
  poverty_lines = read_tendulkar_poverty_lines(),
  reference_period = seq(as.Date("2011-07-01"), as.Date("2012-06-01"), by = "month"),
  reference_index = NULL,
  start_period = NULL,
  end_period = NULL,

```

```

    reference_rupees = tendulkar_real_poverty_line(),
    require_complete_grid = TRUE) {
temporal_index <- if (inherits(temporal_series, "emi_temporal_price_series")) {
  temporal_series$index
} else {
  safe_df(temporal_series)
}
expanded <- apply_price_state_rules(
  temporal_index,
  state_rules,
  start_period = start_period,
  end_period = end_period,
  require_complete_grid = require_complete_grid
)
spatial <- build_tendulkar_spatial_relatives(poverty_lines, reference_rupees)
missing_spatial <- setdiff(
  unique(paste(expanded$state_code, expanded$sector, sep = "\r")),
  unique(paste(spatial$state_code, spatial$sector, sep = "\r"))
)
if (length(missing_spatial)) {
  stop("Tendulkar poverty-line data do not cover every state and rural/urban CPI series.", call. = FALSE)
}
build_state_sector_deflator(
  expanded,
  spatial,
  reference_period = reference_period,
  reference_index = reference_index
)
}

```

Output: outputs/figures/appendix/appendix_consumption_hces_consistency.pdf

Rendering code: R/output/appendix_data_construction_exhibits.R and R/output/save_figures.R

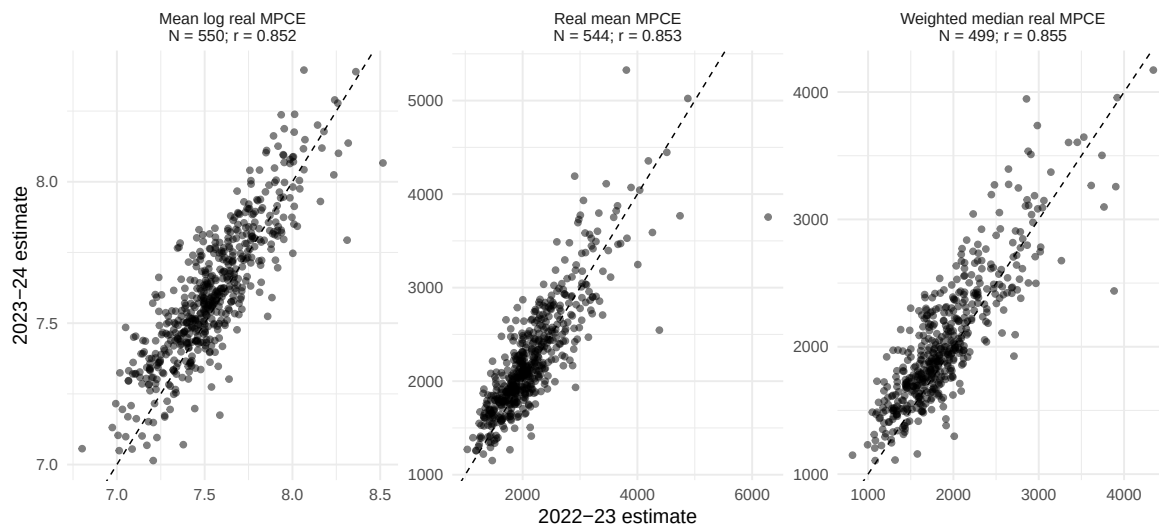


Figure 8: Agreement between district real-consumption estimates in the 2022-23 and 2023-24 HCES rounds. Each panel uses districts retained in both rounds under the consumption sample criteria; the dashed line is the 45-degree line. Sources: National Sample Survey Office, HCES 2022-23 and HCES 2023-24 (National Sample Survey Office 2024, 2025).

Testing CPI Linking Across Index Bases

Small rural and urban CPI examples have link factors known in advance. The first test verifies that each link factor equals the median monthly ratio of the new index to the old index and that the overlap count is retained. The second verifies that construction stops when a historical series has fewer common months with CPI-R/U than the required minimum.

File: `tests/testthat/test-prices-and-controls.R`

```
test_that("CPI link factors equal median overlap ratios", {
  months <- as.Date(c("2013-01-01", "2013-02-01", "2013-03-01"))
  old <- rbind(
    data.frame(state_code = "A", sector = "rural", period = months, index = c(100, 110, 120)),
    data.frame(state_code = "A", sector = "urban", period = months, index = c(200, 220, 240))
  )
  new <- rbind(
    data.frame(state_code = "A", sector = "rural", period = months, index = c(200, 220, 360)),
    data.frame(state_code = "A", sector = "urban", period = months, index = c(100, 110, 120))
  )
  links <- summarise_price_links(old, new, months[1], months[3])
  expect_equal(links$link_factor, c(2, 0.5))
  expect_equal(links$link_factor_first, c(2, 0.5))
})
```

```

expect_equal(links$link_factor_last, c(3, 0.5))
expect_equal(links$link_months, c(3L, 3L))
})

test_that("temporal price construction requires enough overlap months", {
  months <- as.Date(c("2012-12-01", "2013-01-01"))
  sources <- list(
    cpi_alrl = data.frame(
      state_code = "A", labour_series = "rural_labour", sector = "rural",
      period = months, index = c(100, 101)
    ),
    cpi_iw_states = data.frame(
      state_code = "A", sector = "urban", period = months, index = c(100, 101)
    ),
    cpi_ruc_2012 = rbind(
      data.frame(state_code = "A", sector = "rural", period = months[2], index = 100),
      data.frame(state_code = "A", sector = "urban", period = months[2], index = 100)
    )
  )
  expect_error(build_temporal_price_series(
    sources,
    overlap_start = months[2], overlap_end = months[2], minimum_link_months = 2
  ))
})

```